LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

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Abstract

Multicenter medical models are typically trained asynchronously across hospitals with heterogeneous acquisition devices, patient populations, and diagnostic-label coverage. We study single-shot asynchronous medical model merging, where each client uploads its locally trained model only once and the server constructs a global model without accessing raw medical images or conducting federated optimization rounds. We identify aggregation-induced prediction collapse as the central failure mode of general-purpose model-merging methods. Furthermore, we show that long-tailed medical priors can make majority-class collapse deceptively competitive in terms of raw accuracy, thereby masking this failure. To address this issue, we propose LAMP-Merge, a long-tail-aware medical prototype-merging framework that reconstructs a diagnostic prototype head from client-side class statistics and calibrates long-tail prevalence using a bounded prior bias. Across five medical datasets, four vision-model families, three client-population sizes, and three levels of non-IID skew, LAMP-Merge matches or surpasses the strongest non-LAMP baseline in 57 of 60 client-averaged settings, while improving balanced accuracy, macro-F1, and prediction-collapse diagnostic metrics.

Index Terms:medical model merging, asynchronous fusion, predictive collapse, long-tail calibration, privacy-preserving learning

I Introduction

Medical artificial intelligence increasingly requires **multi-center collaboration** because no single hospital can cover the full disease spectrum, rare cases, and imaging conditions (Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021). Yet privacy requirements, device heterogeneity, and institutional workflows make **centralizing raw images impractical** (Kaissis et al. 2020a; Bonawitz et al. 2019; Abadi et al. 2016). A realistic alternative is for hospitals to train locally and share limited post-training information.

This setting requires **single-shot asynchronous model merging**. Hospitals may complete training at different times and cannot necessarily participate in repeated global updates, as assumed by conventional federated and decentralized training protocols(McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Warnat-Herresthal et al. 2021). The server must therefore construct one deployable diagnostic model from **exactly one upload per client**, without accessing raw medical images.

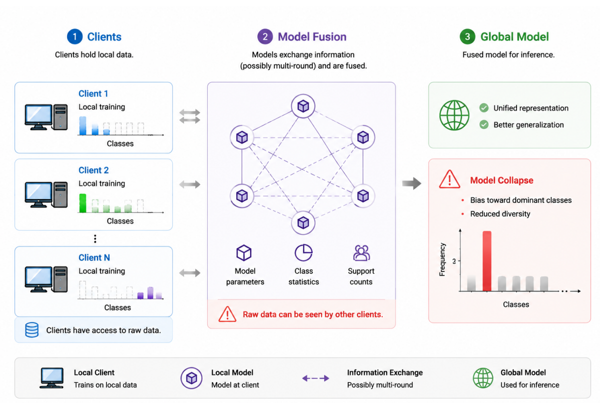

Figure 1: In multicenter healthcare settings, individual clients typically observe only a subset of the global diagnostic space. Conventional parameter-level model-merging methods can therefore cause the merged global model's predictions to concentrate on a small number of dominant classes.

Medical imaging makes the problem especially difficult because hospitals differ not only in device domain but also in **diagnostic-class availability** (Guan and Liu 2022; Zech et al. 2018; Li et al. 2021). Local models are partial experts: a client may provide strong evidence for common diagnoses while lacking rare ones. Moreover, medical labels are often **long-tailed**, so a useful merge must preserve distributed class-specific evidence while respecting genuine prevalence (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

We propose **LAMP-Merge**, a long-tail-aware medical prototype-merging framework. It recasts global parameter averaging as **class-level diagnostic-evidence reconstruction**: each client uploads class prototypes and counts computed with a shared reference extractor, and the server constructs a global prototype classifier with bounded prevalence calibration. The procedure requires one upload and never exposes raw images, per-sample features, logits, or predictions.

Our main contributions are as follows:

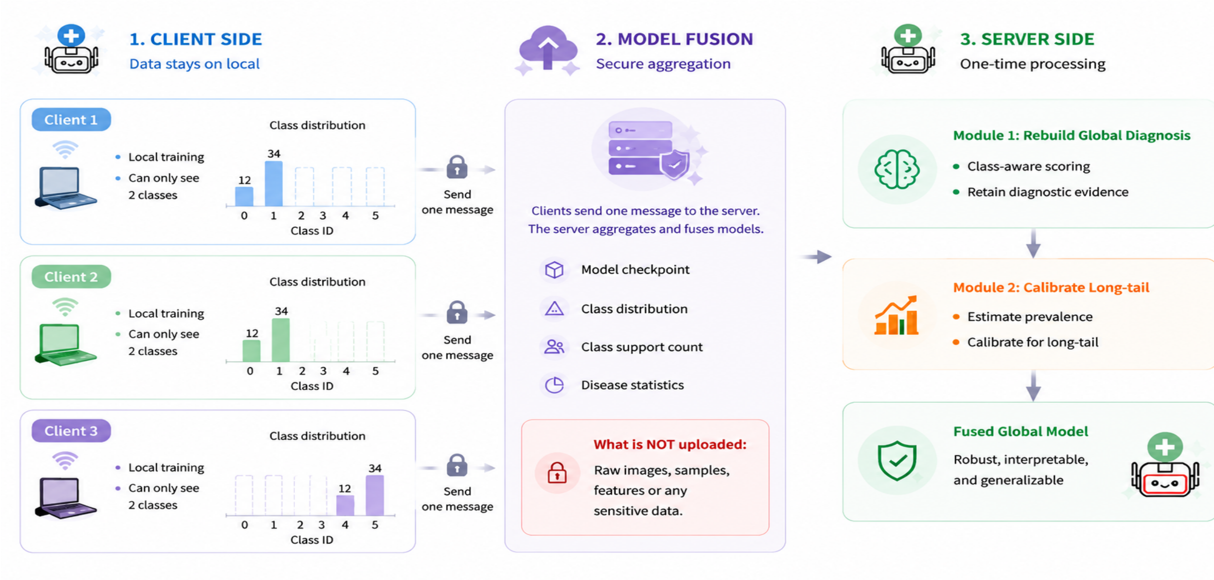

Figure 2: Local models from multicenter hospitals typically contain only partial diagnostic knowledge, while raw medical images cannot be centrally shared. LAMP-Merge reconstructs a privacy-friendly global medical classifier through diagnostic prototype reconstruction and long-tail prevalence calibration, without accessing raw data or requiring multi-round federated optimization.

- We formalize **single-shot asynchronous medical image model merging**, where independently trained clients provide one upload for global model construction.
- We identify **heterogeneous diagnostic-class availability and long-tailed disease prevalence** as the defining structures of this setting.
- We propose **LAMP-Merge**, which reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics to mitigate prediction collapse without exposing raw images.

II Related Works

Model Merging

Model-merging methods commonly integrate checkpoints in parameter, task-vector, or geometric spaces. Weight averaging and Model Soups interpolate model parameters(Izmailov et al. 2018; Wortsman et al. 2022); permutation-aware matching aligns hidden-unit symmetries before averaging(Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023); ZipIt performs feature-level zipping across models(Stoica et al. 2024); Task Arithmetic, TIES-Merging, and DARE operate on task deltas and sign conflicts(Ilharco et al. 2023; Yadav et al. 2023; Yu et al. 2024); Fisher merging, RegMean, Model Stock, Breadcrumbs, Iso-C, FreeMerge, and RobustMerge further introduce curvature, mean-statistic, subspace, or robustness assumptions(Matena and Raffel 2022; Jin et al. 2025; Jang, Yun, and Han 2025; Davari and Belilovsky 2024; Marczak et al. 2025; Zheng and Wang 2025; Zeng et al. 2025).

However, these variables are still global parameters, task vectors, or update directions rather than class-level diagnostic evidence. **They do not ask whether a client has reliable supervision for a specific medical class.** Under incomplete local class coverage and long-tailed prevalence, reliable directions from clients that observed a class can be averaged with absent or conflicting directions from clients that did not, leading to aggregation-induced prediction collapse(Buda, Maki, and Mazurowski 2018; Zech et al. 2018).

Medical Model Merging

Medical fusion studies often exploit domain structures such as irregular clinical time series, physiological sensor graphs, and image-EHR fusion. mTAN, Raindrop, and MedFuse show that medical tasks benefit from modeling clinical sampling mechanisms, modality heterogeneity, and patient-level temporal relations(Shukla and Marlin 2021; Zhang et al. 2022; Hayat, Geras, and Shamout 2023).

However, these methods usually fuse patient-level inputs or clinical representations rather than independently trained image-classifier checkpoints, and they often require patient samples, timestamps, multimodal records, electronic health records, or server-side validation feedback, which conflicts with common medical privacy constraints(Kaissis et al. 2020a; Sheller et al. 2020; Bonawitz et al. 2019). Med-Merge is a relevant training-free medical knowledge-transfer framework(Wei et al. 2026); however, it targets cross-modal adaptation for medical vision-language models rather than asynchronous checkpoint merging for the same classification task. **Thus, existing medical fusion routes are mismatched with our privacy and single-upload protocol constraints.** LAMP-Merge keeps the medical structure uploadable by using only class-level prototypes and prevalence counts computed locally by clients.

Federated Learning

Federated and decentralized learning perform collaborative training through communication protocols. FedAvg and its medical deployments rely on repeated server-client updates(McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Li et al. 2021; Kaissis et al. 2020b; Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021); Swarm Learning and Gossip Learning reduce centralization but still require participants to continue training, exchanging models, and optimizing from intermediate states(Warnat-Herresthal et al. 2021; Heged\Hus, Danner, and Jelasity 2019).

However, our problem is single-shot asynchronous model merging after local training has finished: hospitals may not remain online, receive global checkpoints repeatedly, or run additional optimization rounds. **The limitation is therefore communication feasibility rather than only privacy.** LAMP-Merge requires one upload of a local model interface and class-level statistics, enabling the server to construct a unified diagnostic model without a multi-round feedback loop.

III Method

Motivation and Overall Idea

Under partial local class coverage, parameter-level merging can turn several non-degenerate local predictors into a globally collapsed predictor. Because medical prevalence is often long-tailed, such collapse can still appear competitive in raw accuracy. A useful merger must therefore recover rare-class discrimination without discarding genuine prevalence.

LAMP-Merge addresses this tension with two components. Diagnostic prototype reconstruction aggregates only class-supported evidence in a shared reference space to recover global multiclass directions. Long-tail prevalence calibration then applies a bounded prior bias estimated from client-side counts, preserving clinically meaningful majority prevalence without replacing those directions. The resulting one-shot protocol requires neither additional federated training nor raw-data exposure.

Overview

The core framework of LAMP-Merge is illustrated in figure2. Its central idea is to recast medical model merging from global parameter averaging into the reconstruction of class-conditional diagnostic evidence, supplemented by class-prior information.

Notably, LAMP-Merge neither requires the server to access clients' raw images nor relies on a server-side validation set, per-sample features, or per-sample predictions. Clients upload only class-level statistics; thus, the framework preserves a single-shot asynchronous communication boundary, distinguishing it from federated learning procedures that require repeated model broadcasts and local retraining.

The framework comprises two key components: a class-level diagnostic prototype reconstruction module and a long-tail prevalence calibration module. The prototype reconstruction module restores multiclass discriminative directions across the complete global diagnostic space, after which the calibration module retains appropriate class-prior information.

Diagnostic Prototype Reconstruction Module

Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c. In medical partitions, $D_{i,c}$ can be empty for many classes. Thus, a client can be reliable for a subset of diagnoses but unreliable for unseen classes. A class-agnostic parameter merge has no explicit mechanism to preserve this local reliability structure.

Therefore, LAMP-Merge first reconstructs a class-wise diagnostic head in the shared reference feature space. Let ϕ_0 be the frozen reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c, client i computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y)\in D_i} \mathbf{1}[y = c]\phi_0(T(x)), \tag{1}$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

The server converts support counts into evidence weights:

$$e_{i,c} = (n_{i,c} + 1)^\gamma \mathbf{1}[n_{i,c} > 0], \tag{2}$$

where $\gamma = 0.45$ is the default evidence exponent. The normalized reliability of client i for class c is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^{K} e_{j,c}}. \tag{3}$$

The global diagnosis prototype is then

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}. \tag{4}$$

Finally, LAMP-Merge synthesizes a cosine prototype classifier:

$$w_c = s\frac{p_c}{\|p_c\|_2}, \tag{5}$$

where $s = 20$ is the default head scale. This module addresses a key limitation of existing methods: it preserves the classes for which each client has supporting evidence and reconstructs a global multiclass diagnostic head.

Long-tail Prevalence Calibration

After obtaining this newly reconstructed diagnostic head, each client additionally uploads class-prevalence counts $m_{i,c}$ to account for the long-tailed prevalence inherent in real-world medical tasks. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^{K} m_{i,c}}{\sum_{k=1}^{C}\sum_{i=1}^{K} m_{i,k}}. \tag{6}$$

We measure dominant-class imbalance by

$$r = C \max_c \pi_c. \tag{7}$$

The module is activated only when $r > \tau$, where $\tau = 2.5$ by default. When activated, it adds a centered log-prior bias

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_{k=1}^{C} \log \pi_k \right), \tag{8}$$

with default maximum calibration strength $\lambda = 5.0$. The final score is

$$\mathrm{score}_c(x) = w_c^\top \phi_0(T(x)) + b_c. \tag{9}$$

The bias is centered, bounded, and attached to the prototype classifier; it calibrates prevalence without replacing the multi-class diagnostic directions reconstructed by the Diagnostic Prototype Reconstruction Module.

Experiments

Experimental Settings

1) Evaluation Datasets: We evaluate the merged models on five representative MedMNIST v2 benchmarks(Yang, Shi, and Ni 2023): Blood, Derma, Organ-C, Organ-S, and Ultrasound. They cover blood-cell microscopy, dermoscopy, abdominal CT, and ultrasound image classification (detailed dataset descriptions are provided in the Appendix).

2) Baselines: We compare LAMP-Merge with twelve model-merging baselines covering the main technical routes of current model merging: `avg`(Wortsman et al. 2022), `ties`(Yadav et al. 2023), `dare_linear`(Yu et al. 2024), `dare_ties`(Yu et al. 2024), `regmean`(Jin et al. 2025), `fisher`(Matena and Raffel 2022), `breadcrumbs`(Davari and Belilovsky 2024), `model_stock`(Jang, Yun, and Han 2025), `from`(Ilharco et al. 2023), `iso_c`(Marczak et al. 2025), `free_merge`(Zheng and Wang 2025), and `robustmerge`(Zeng et al. 2025). These baselines include weight averaging, task-vector pruning, sign-conflict resolution, curvature or mean-statistic correction, stock-style model interpolation, isotropic subspace merging, Fourier-domain fusion, and direction-robust merging.

3) Merging Settings: Experiments follow a single-shot asynchronous model-merging protocol. Each client first independently trains a local model on its own medical imaging data, then computes the class-level statistics required by LAMP-Merge locally, including class-support counts, reference-space class prototypes, and class-prevalence counts. These statistics are uploaded once together with an interface to the local checkpoint. The server subsequently performs a single model-merging step using only the uploaded checkpoints and class-level statistics, and evaluates the resulting merged model on a global test set.

We evaluate four vision-model families: ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny(He et al. 2016; Liu et al. 2022; Dosovitskiy et al. 2021; Liu et al. 2021). For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$, using seed 42(Hsu, Qi, and Brown 2019). A client-average result fixes the dataset, model, and K, and then averages over the three β values.

4) Implementation Environment: All experiments were run on a server with two Intel(R) Xeon(R) Silver 4210 CPUs, four NVIDIA GeForce RTX 2080 Ti GPUs, and the conda environment `MM` with Python 3.10.20, PyTorch 2.6.0+`cu118`, and CUDA runtime 11.8.

Comparison With the State of the Arts

The main comparison uses client-average ACC: for each fixed dataset, model, and client number, we first average over the three β skew levels and then compare merging methods. A cell is counted as successful when LAMP-Merge is greater than or equal to the strongest non-LAMP baseline in that cell.

Tables 1–4 report the full client-average results for all four backbones. Across the 60 client-average cells, LAMP-Merge wins or ties the strongest non-LAMP baseline in **57 cells (95.0%)**.

Ablation Study

Module-level ablations use exactly the same experimental setup as the main experiments. We compare three configurations: full LAMP-Merge, diagnostic prototype reconstruction only, and weight averaging augmented only with long-tail prevalence calibration. Both components provide substantial performance improvements.

1) Effectiveness of Diagnostic Prototype Reconstruction To further identify which information within diagnostic prototype reconstruction is necessary, we conduct controlled substitutions across the full setting. All configurations retain long-tail prevalence calibration and vary only the prototype semantics or support-count weighting.

Fig. 3 and Table 5 show that neither an arbitrary head nor a class-agnostic statistic explains the gain. Replacing reference-space prototypes or class-specific support weights consistently reduces ACC, confirming that **diagnostic semantics and class-level reliability are jointly necessary**.

2) Effectiveness of Long-tail Prevalence Calibration We further replace the prior term inside long-tail prevalence calibration. This group changes only the class prior used by the calibration bias.

Table 6 shows that removing prevalence calibration or replacing the real prior with a uniform prior reduces mean ACC, confirming that long-tailed prevalence is part of the full method. The smoothed prior remains close to the official method, indicating that LAMP-Merge does not depend on accidental fluctuations of exact counts; bounded calibration remains effective as long as the prior preserves the real long-tailed structure.

New-metric Diagnostic Experiment

Although the main experiments use ACC for direct comparison, long-tailed medical distributions can allow single-class or few-class predictors to achieve high ACC(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). We therefore report balanced accuracy, macro-F1, collapse ratio, effective predicted classes, and predicted–true total variation.

Table 7 compares LAMP-Merge with the best-performing baseline for each metric. **LAMP-Merge consistently and**

Table 1: Client-average test accuracy on ResNet. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

	Blood				Derma				Organ-C				Organ-S				Ultrasound			
Method	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**
avg(Wortsman et al. 2022)	0.3345	0.2467	0.2822	0.2878	0.6291	0.4700	0.6244	0.5745	0.2261	0.1772	0.1396	0.1810	0.2864	0.1794	0.2475	0.2378	0.2435	0.1581	0.1542	0.1853
ties(Yadav et al. 2023)	0.2601	0.1872	0.1787	0.2087	0.5995	0.4841	0.6484	0.5773	0.2650	0.1684	0.2118	0.2151	0.1981	0.1915	0.1874	0.1923	0.3477	0.1989	0.1854	0.2440
dare_linear(Yu et al. 2024)	0.2832	0.2856	0.2392	0.2693	0.6357	0.4841	0.6288	0.5829	0.2018	0.2588	0.1650	0.2085	0.2545	0.2245	0.2537	0.2442	0.2162	0.1728	0.1608	0.1833
dare_ties(Yu et al. 2024)	0.3194	0.1935	0.1780	0.2303	0.5375	0.4825	0.5558	0.5253	0.1851	0.1515	0.1859	0.1742	0.1790	0.1352	0.1730	0.1624	0.2932	0.1635	0.1890	0.2152
regmean(Jin et al. 2025)	0.3460	0.2369	0.3407	0.3079	0.5812	0.2843	0.5022	0.4559	0.2405	0.1340	0.1388	0.1711	0.2504	0.1623	0.1493	0.1873	0.2276	0.1695	0.1318	0.1763
fisher(Matena and Raffel 2022)	0.3196	0.2237	0.2608	0.2680	0.6707	0.6702	0.6690	0.6700	0.2973	0.1741	0.1812	0.2175	0.2010	0.2956	0.1963	0.2310	0.2333	0.1860	0.2513	0.2235
breadcrumbs(Davari and Belilovsky 2024)	0.0854	0.1417	0.1460	0.1244	0.0495	0.2913	0.1721	0.1710	0.0766	0.0676	0.0685	0.0709	0.0802	0.0948	0.0603	0.0784	0.1225	0.1273	0.1021	0.1173
model_stock(Jang, Yun, and Han 2025)	0.2420	0.1824	0.1975	0.2073	0.6216	0.4311	0.5721	0.5416	0.1225	0.1274	0.1185	0.1228	0.2742	0.1578	0.2012	0.2111	0.1791	0.1321	0.1578	0.1563
from(Ilharco et al. 2023)	0.2989	0.2306	0.2052	0.2449	0.6713	0.4369	0.6580	0.5887	0.2350	0.1560	0.1520	0.1810	0.2207	0.2087	0.2476	0.2257	0.2309	0.1506	0.2063	0.1959
iso_c(Marczak et al. 2025)	0.3278	0.2748	0.3023	0.3016	0.5049	0.4422	0.6416	0.5296	0.2331	0.1629	0.1383	0.1781	0.2736	0.2118	0.2453	0.2436	0.2615	0.1698	0.1554	0.1956
free_merge(Zheng and Wang 2025)	0.3345	0.2467	0.2822	0.2878	0.6291	0.4700	0.6246	0.5746	0.2261	0.1771	0.1396	0.1809	0.2863	0.1793	0.2475	0.2377	0.2435	0.1581	0.1545	0.1854
robustmerge(Zeng et al. 2025)	0.3401	0.2430	0.2256	0.2696	0.2956	0.2085	0.3942	0.2994	0.2458	0.1764	0.1653	0.1958	0.2680	0.2227	0.2301	0.2403	0.2189	0.2231	0.1707	0.2042
LAMP-Merge	0.8002	0.7999	0.8007	0.8003	0.6745	0.6761	0.6763	0.6756	0.6074	0.6069	0.6076	0.6073	0.5531	0.5524	0.5517	0.5524	0.4762	0.4762	0.4762	0.4762

Table 2: Client-average test accuracy on ConvNeXt. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

	Blood				Derma				Organ-C				Organ-S				Ultrasound			
Method	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**
avg(Wortsman et al. 2022)	0.1565	0.1122	0.1384	0.1357	0.2966	0.4830	0.6688	0.4828	0.1037	0.1218	0.0944	0.1066	0.1308	0.1035	0.1416	0.1253	0.1506	0.1506	0.1252	0.1421
ties(Yadav et al. 2023)	0.1715	0.1214	0.1524	0.1484	0.2966	0.1797	0.2713	0.2492	0.1217	0.0786	0.0716	0.0906	0.1123	0.0761	0.0796	0.0893	0.1518	0.1456	0.1291	0.1422
dare_linear(Yu et al. 2024)	0.1756	0.1122	0.1345	0.1408	0.2705	0.4830	0.4830	0.4122	0.1035	0.0699	0.1085	0.0940	0.1282	0.1035	0.0797	0.1038	0.1575	0.1575	0.1264	0.1471
dare_ties(Yu et al. 2024)	0.1096	0.1519	0.1497	0.1371	0.0908	0.2798	0.2451	0.2052	0.1305	0.1346	0.0770	0.1140	0.1096	0.0946	0.0709	0.0917	0.1456	0.1297	0.1500	0.1418
regmean(Jin et al. 2025)	0.1565	0.0817	0.1856	0.1413	0.2633	0.0773	0.2720	0.2042	0.1143	0.1017	0.0768	0.0976	0.1014	0.1259	0.1038	0.1104	0.1575	0.1539	0.1500	0.1538
fisher(Matena and Raffel 2022)	0.1565	0.0999	0.1189	0.1251	0.4825	0.2638	0.2971	0.3478	0.1003	0.0974	0.0915	0.0964	0.0604	0.0517	0.0843	0.0655	0.1695	0.1360	0.1456	0.1504
breadcrumbs(Davari and Belilovsky 2024)	0.1516	0.1366	0.1163	0.1348	0.0524	0.2766	0.4630	0.2640	0.1031	0.1444	0.1574	0.1350	0.1516	0.1554	0.2077	0.1716	0.1384	0.1363	0.1363	0.1370
model_stock(Jang, Yun, and Han 2025)	0.1756	0.1144	0.1778	0.1559	0.2966	0.4830	0.6688	0.4828	0.1276	0.1218	0.1385	0.1293	0.1308	0.1035	0.1935	0.1426	0.1575	0.1291	0.1479	0.1448
from(Ilharco et al. 2023)	0.1565	0.1083	0.1671	0.1440	0.2966	0.3406	0.4569	0.3647	0.1232	0.1224	0.0650	0.1035	0.1324	0.1105	0.0897	0.1109	0.1575	0.1578	0.1087	0.1413
iso_c(Marczak et al. 2025)	0.1947	0.1865	0.1480	0.1764	0.2971	0.4830	0.6688	0.4830	0.0991	0.0699	0.0945	0.0878	0.0789	0.1477	0.0867	0.1044	0.1518	0.1291	0.1605	0.1471
free_merge(Zheng and Wang 2025)	0.1575	0.1190	0.1384	0.1383	0.2971	0.4830	0.6688	0.4830	0.0906	0.1218	0.1385	0.1170	0.1308	0.1035	0.1416	0.1253	0.1506	0.1291	0.1695	0.1497
robustmerge(Zeng et al. 2025)	0.1410	0.1405	0.0752	0.1189	0.0908	0.4507	0.4830	0.3415	0.1394	0.1218	0.1793	0.1468	0.2354	0.0961	0.2077	0.1797	0.1300	0.1135	0.1363	0.1266
LAMP-Merge	0.8482	0.8483	0.8493	0.8486	0.5992	0.5920	0.5852	0.5921	0.6482	0.6480	0.6495	0.6486	0.5977	0.5985	0.5990	0.5984	0.4259	0.4259	0.4259	0.4259

Table 3: Client-average test accuracy on ViT-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

	Blood				Derma				Organ-C				Organ-S				Ultrasound			
Method	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**
avg(Wortsman et al. 2022)	0.1343	0.1082	0.1160	0.1195	0.4638	0.4369	0.3696	0.4234	0.1657	0.1043	0.0952	0.1217	0.0844	0.0852	0.1710	0.1135	0.1450	0.1327	0.1884	0.1554
ties(Yadav et al. 2023)	0.0796	0.1421	0.1285	0.1167	0.4780	0.3498	0.4615	0.4298	0.1341	0.1223	0.0953	0.1172	0.1192	0.1002	0.1597	0.1264	0.2001	0.1306	0.1018	0.1442
dare_linear(Yu et al. 2024)	0.2397	0.1129	0.1387	0.1638	0.2850	0.1002	0.1204	0.1685	0.1534	0.0946	0.0704	0.1061	0.0748	0.0549	0.1611	0.0969	0.1366	0.1051	0.1435	0.1284
dare_ties(Yu et al. 2024)	0.0735	0.1189	0.1146	0.1023	0.2735	0.4502	0.3528	0.3588	0.1252	0.0903	0.1033	0.1063	0.0541	0.0874	0.0591	0.0669	0.1447	0.1120	0.1042	0.1203
regmean(Jin et al. 2025)	0.1748	0.1417	0.1290	0.1485	0.5114	0.3667	0.6658	0.5146	0.1250	0.0964	0.0983	0.1066	0.1263	0.0717	0.1068	0.1016	0.1491	0.1384	0.1144	0.1340
fisher(Matena and Raffel 2022)	0.1166	0.1512	0.1944	0.1541	0.4780	0.2042	0.4820	0.3881	0.0799	0.1072	0.1424	0.1098	0.0856	0.1218	0.1222	0.1099	0.1611	0.1315	0.1420	0.1449
breadcrumbs(Davari and Belilovsky 2024)	0.1363	0.1039	0.1300	0.1234	0.0489	0.2572	0.0607	0.1223	0.0773	0.1000	0.1644	0.1139	0.1237	0.0483	0.1219	0.0980	0.1360	0.1141	0.1524	0.1342
model_stock(Jang, Yun, and Han 2025)	0.1940	0.1120	0.1134	0.1398	0.3805	0.3079	0.2735	0.3206	0.1488	0.1227	0.1215	0.1310	0.0893	0.0859	0.1867	0.1206	0.1599	0.1809	0.1641	0.1683
from(Ilharco et al. 2023)	0.1802	0.0781	0.1405	0.1329	0.4695	0.5121	0.6688	0.5501	0.2148	0.1301	0.0972	0.1474	0.0772	0.0798	0.1485	0.1018	0.1836	0.1743	0.1776	0.1785
iso_c(Marczak et al. 2025)	0.1234	0.1483	0.1018	0.1245	0.5709	0.4529	0.2306	0.4181	0.1084	0.0567	0.0897	0.0849	0.0611	0.0531	0.1081	0.0741	0.1417	0.1527	0.1869	0.1604
free_merge(Zheng and Wang 2025)	0.1989	0.1457	0.1392	0.1613	0.4630	0.4770	0.3855	0.4418	0.1425	0.0980	0.0759	0.1055	0.0727	0.0914	0.1643	0.1095	0.1405	0.1593	0.1647	0.1548
robustmerge(Zeng et al. 2025)	0.1482	0.1070	0.1425	0.1326	0.0347	0.2768	0.1724	0.1613	0.0954	0.0617	0.1438	0.1003	0.1260	0.0707	0.1217	0.1061	0.1698	0.1854	0.1563	0.1705
LAMP-Merge	0.8523	0.8522	0.8516	0.8520	0.5970	0.5938	0.5860	0.5923	0.5805	0.5790	0.5790	0.5795	0.5406	0.5372	0.5388	0.5389	0.4645	0.4645	0.4645	0.4645

Table 4: Client-average test accuracy on Swin-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

	Blood				Derma				Organ-C				Organ-S				Ultrasound			
Method	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**	$K=3$	$K=5$	$K=7$	**Avg**
avg(Wortsman et al. 2022)	0.1565	0.1186	0.1524	0.1425	0.4569	0.4830	0.6688	0.5362	0.1068	0.1294	0.0787	0.1050	0.0682	0.0604	0.1381	0.0889	0.1527	0.1294	0.1465	0.1429
ties(Yadav et al. 2023)	0.0972	0.1190	0.1384	0.1182	0.0841	0.0780	0.6688	0.2770	0.1160	0.0702	0.0934	0.0932	0.0734	0.0607	0.1107	0.0816	0.1695	0.1303	0.1261	0.1420
dare_linear(Yu et al. 2024)	0.1565	0.1161	0.1825	0.1517	0.4569	0.2771	0.4630	0.3990	0.1155	0.1272	0.0776	0.1068	0.0640	0.0720	0.1381	0.0914	0.1743	0.1536	0.1497	0.1592
dare_ties(Yu et al. 2024)	0.1384	0.1162	0.1384	0.1310	0.2961	0.2638	0.6688	0.4096	0.1276	0.0774	0.1116	0.1055	0.1194	0.0434	0.1230	0.0953	0.1791	0.1351	0.1300	0.1481
regmean(Jin et al. 2025)	0.1374	0.1715	0.1578	0.1556	0.2705	0.4630	0.4569	0.3968	0.1411	0.0773	0.0811	0.0998	0.1215	0.0821	0.0926	0.0987	0.1465	0.1105	0.1779	0.1450
fisher(Matena and Raffel 2022)	0.1633	0.1756	0.1565	0.1651	0.4582	0.0652	0.4718	0.3317	0.0814	0.0899	0.0978	0.0897	0.0651	0.0650	0.1092	0.0798	0.1746	0.1165	0.1755	0.1555
breadcrumbs(Davari and Belilovsky 2024)	0.1500	0.1365	0.1537	0.1467	0.0647	0.2771	0.0715	0.1378	0.1445	0.1783	0.1660	0.1629	0.2354	0.2077	0.2111	0.2181	0.1450	0.1312	0.1653	0.1472
model_stock(Jang, Yun, and Han 2025)	0.1374	0.1440	0.1715	0.1510	0.4569	0.2771	0.4825	0.4055	0.1110	0.0771	0.1404	0.1095	0.0671	0.1106	0.1714	0.1164	0.1539	0.1273	0.1869	0.1560
from(Ilharco et al. 2023)	0.1565	0.1495	0.1671	0.1577	0.4569	0.2971	0.6688	0.4743	0.1174	0.1220	0.1051	0.1148	0.1133	0.1016	0.1380	0.1176	0.1539	0.1252	0.1476	0.1422
iso_c(Marczak et al. 2025)	0.1430	0.1722	0.1700	0.1617	0.3099	0.2840	0.5174	0.3704	0.1302	0.1327	0.1469	0.1366	0.1545	0.1727	0.1880	0.1717	0.1755	0.1731	0.1791	0.1759
free_merge(Zheng and Wang 2025)	0.1154	0.0861	0.1840	0.1285	0.4569	0.4830	0.6688	0.5362	0.0959	0.1377	0.0890	0.1075	0.0682	0.0497	0.1062	0.0747	0.1518	0.1575	0.1599	0.1564
robustmerge(Zeng et al. 2025)	0.1374	0.1409	0.1670	0.1484	0.2377	0.2771	0.0504	0.1884	0.1601	0.1442	0.1960	0.1668	0.1673	0.1151	0.2077	0.1634	0.1761	0.1315	0.1629	0.1568
LAMP-Merge	0.7737	0.7753	0.7748	0.7746	0.6374	0.6251	0.6244	0.6290	0.6651	0.6632	0.6630	0.6638	0.6078	0.6076	0.6068	0.6074	0.4816	0.4816	0.4816	0.4816

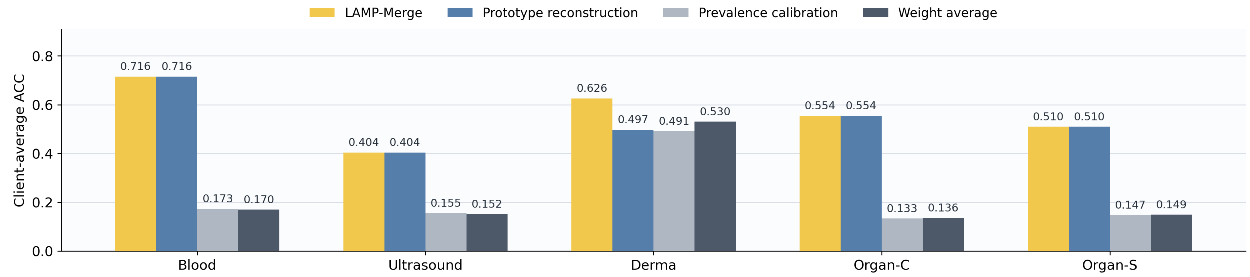

Figure 3: Dataset-level client-average accuracy in the module ablation. Each dataset contains 12 cells from four vision-model families and three client counts; each cell is averaged over three Dirichlet skew levels.

Table 5: Internal ablation of diagnostic prototype reconstruction. All settings cover 180 raw cells and 60 client-average cells.

Setting	Mean Acc	Δ vs. LAMP
LAMP-Merge	0.6209	–
Classifier-head aggregation	0.2579	↓0.3630
Global-feature mean	0.2645	↓0.3564
Support-only synthetic head	0.1137	↓0.5073
Shuffled-label prototype	0.2002	↓0.4207
Uniform client weight	0.5839	↓0.0370
Binary support only	0.5517	↓0.0692
Global client-size weight	0.5944	↓0.0266

Table 6: Internal ablation of long-tail prevalence calibration.

Setting	Mean Acc	Δ vs. LAMP
LAMP-Merge	0.6209	–
No prevalence calibration	0.5879	↓0.0330
Uniform prevalence prior	0.5879	↓0.0330
Smoothed prevalence prior	0.6209	0.0000

substantially outperforms the existing methods. This indicates that its accuracy gains do not arise from more severe single-class prediction collapse, but rather from more complete global diagnostic discrimination.

Visualization Experiment

To visualize the prediction-distribution changes measured by the new diagnostic metrics, we compare the predicted class allocation induced by LAMP-Merge with that of the strongest generic merge. The generic merge concentrates predictions on one class, whereas LAMP-Merge recovers a broader class allocation aligned with the test distribution.

Hyperparameter Analysis Experiment

Fig. 5 (top) evaluates the prototype-head scale s and calibration strength λ over 180 raw cells and 60 client-average cells. The default $s = 20$ and $\lambda = 5$ lie in broad stable regions, indicating that the gain is not produced by narrow hyperparameter tuning.

t-SNE Visualization Experiment

Fig. 5 (middle) uses t-SNE(van der Maaten and Hinton 2008) only for analysis on Blood with ResNet, $K = 3$,

Table 7: Prediction-collapse diagnostics averaged over five medical datasets and four backbones per dataset.

Metric	LAMP-Merge	Best reference	Δ vs. LAMP
Accuracy ↑	**0.5885**	0.3156	↑+0.2729
BA / macro recall ↑	**0.5747**	0.1810	↑+0.3937
Macro F1 ↑	**0.5352**	0.1123	↑+0.4229
Collapse ratio ↓	**0.2294**	0.7477	↓-0.5184
Effective classes ↑	**8.0390**	2.1360	↑+5.9030
Pred-true TV ↓	**0.1510**	0.6168	↓-0.4658

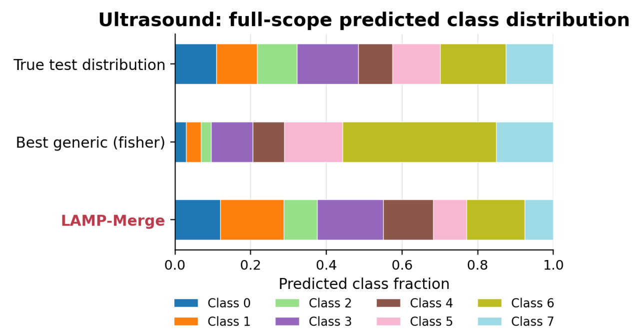

Figure 4: Predicted class distributions on Ultrasound, averaged over the four visual backbones, $K \in \{3, 5, 7\}$, and the three Dirichlet skew levels. The generic merge concentrates predictions on one class, while LAMP-Merge recovers a broader class allocation aligned with the test distribution.

and $\beta = 0.01$. LAMP-Merge retains separated output-probability groups, while generic merges exhibit compressed or entangled predictive representations.

In-depth Analysis Experiment

Fig. 5 (bottom) summarizes inter-class separation, nearest-class margin, within-class client consistency, and prototype–client alignment. Together with Table 5, the joint profile confirms that class-semantic reference-space prototypes and class-specific support weights provide the mechanism behind the full-scope gain.

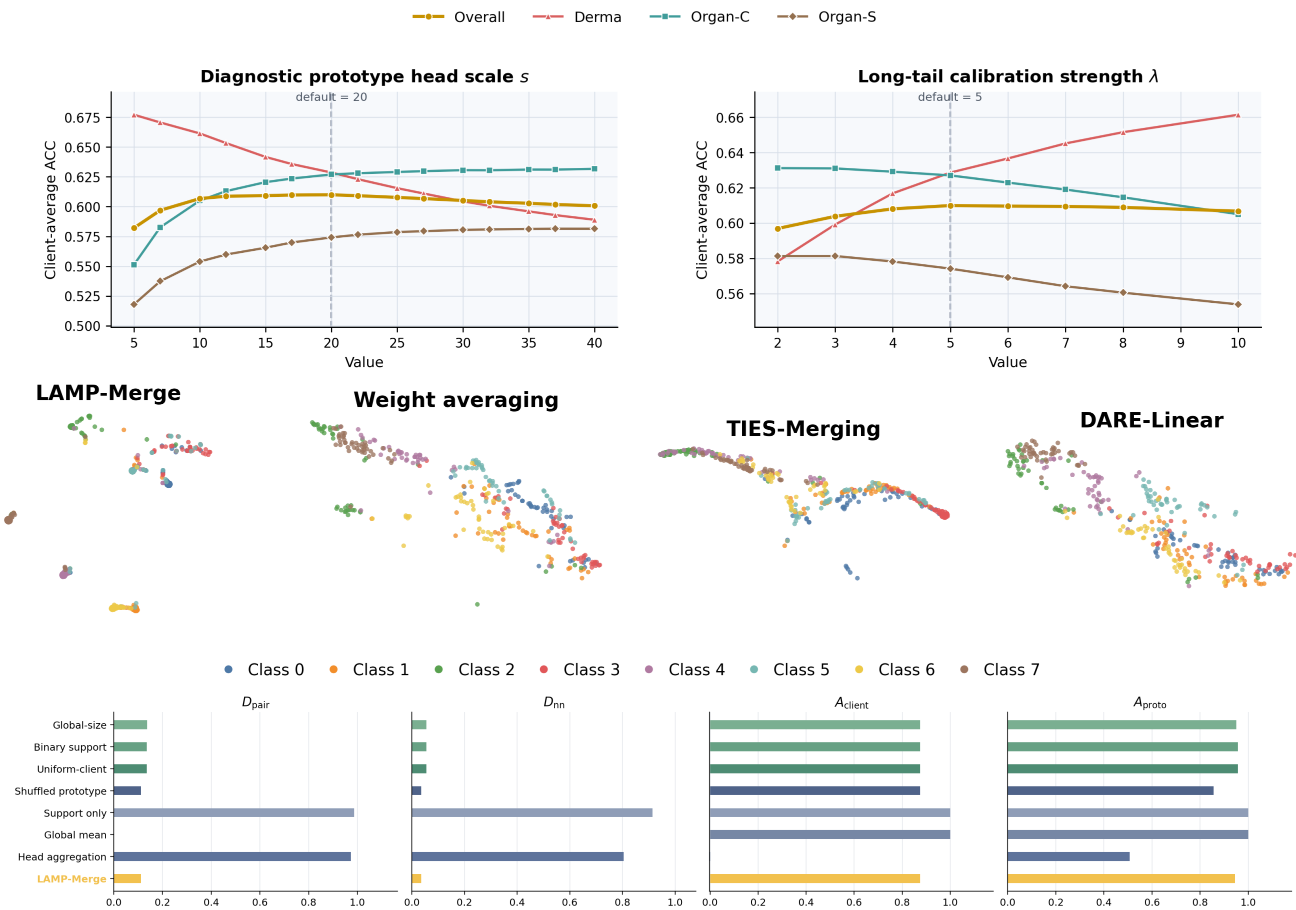

Figure 5: Core mechanism analyses. Top: full-scope hyperparameter sensitivity, where $s = 20$ and $\lambda = 5$ lie in stable high-accuracy regions across 180 raw cells and 60 client-average cells. Middle: joint t-SNE of test-image output-probability representations on Blood; from left to right, LAMP-Merge, weight averaging, TIES-Merging, and DARE-Linear. Bottom: full-scope prototype geometry, summarizing pairwise distance, nearest-class distance, prototype consistency, and prototype–client alignment across controlled prototype substitutions. The three panels jointly show that the performance gain is stable, preserves separated multi-class predictions, and is supported by class-semantic prototype geometry rather than an arbitrary classifier head.

Conclusion

We studied **one-shot asynchronous medical model merging**, where independently trained hospital models must be fused without federated optimization rounds and without exposing raw medical images to the server. Our empirical analysis shows that generic model-merging methods suffer from **aggregation-induced predictive collapse** under incomplete local diagnostic coverage. We further show that, on long-tailed medical datasets, such collapse can be masked by raw accuracy. We therefore introduce the Long-tail Prevalence Calibration module and perform secondary evaluation using macro recall, macro-F1, and predictive-distribution diagnostics(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

Taken together, the full-scope comparisons, prediction-distribution diagnostics, and prototype-error analysis establish a consistent mechanism: class-level support-weighted evidence reconstructs discriminative directions for locally absent diagnoses, while bounded prevalence calibration retains clinically meaningful imbalance without reintroducing majority-class collapse.

LAMP-Merge addresses these two properties through a concise two-module design. Future work will extend its applicability in two directions: first, from medical image classification to segmentation, detection, and multimodal diagnostic tasks; and second, toward stronger privacy-preserving mechanisms, such as differential privacy and secure aggregation, to further reduce potential privacy risks associated with class-level statistics(Kaissis et al. 2020a; Geyer, Klein, and Nabi 2017; Abadi et al. 2016). We will also explore deployment and validation in larger-scale real-world multicenter healthcare settings.

References

Abadi, M.; Chu, A.; Goodfellow, I.; McMahan, H. B.; Mironov, I.; Talwar, K.; and Zhang, L. 2016. Deep Learning with Differential Privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 308–318.

Ainsworth, S. K.; Hayase, J.; and Srinivasa, S. 2023. Git Re-Basin: Merging Models modulo Permutation Symmetries. arXiv:2209.04836.

Bonawitz, K.; et al. 2019. Towards Federated Learning at Scale: System Design. In *Proceedings of Machine Learning and Systems*, volume 1, 374–388.

Brodersen, K. H.; Ong, C. S.; Stephan, K. E.; and Buhmann, J. M. 2010. The Balanced Accuracy and Its Posterior Distribution. In *2010 20th International Conference on Pattern Recognition*, 3121–3124.

Buda, M.; Maki, A.; and Mazurowski, M. A. 2018. A systematic study of the class imbalance problem in convolutional neural networks. *Neural Networks*, 106: 249–259.

Davari, M.; and Belilovsky, E. 2024. Model Breadcrumbs: Scaling Multi-Task Model Merging with Sparse Masks. arXiv:2312.06795.

Dosovitskiy, A.; et al. 2021. An Image Is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations*.

Entezari, R.; Sedghi, H.; Saukh, O.; and Neyshabur, B. 2022. The Role of Permutation Invariance in Linear Mode Connectivity of Neural Networks. In *International Conference on Learning Representations*.

Geyer, R. C.; Klein, T.; and Nabi, M. 2017. Differentially Private Federated Learning: A Client Level Perspective. arXiv:1712.07557.

Guan, H.; and Liu, M. 2022. Domain Adaptation for Medical Image Analysis: A Survey. *IEEE Transactions on Biomedical Engineering*, 69(3): 1173–1185.

Hayat, N.; Geras, K. J.; and Shamout, F. E. 2023. MedFuse: Multi-modal fusion with clinical time-series data and chest X-ray images. arXiv:2207.07027.

He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.

Heged\Hus, I.; Danner, G.; and Jelasity, M. 2019. Gossip Learning as a Decentralized Alternative to Federated Learning. In Pereira, J.; and Ricci, L., eds., *Distributed Applications and Interoperable Systems*, 74–90. Cham: Springer International Publishing. ISBN 978-3-030-22496-7.

Hsu, T.-M. H.; Qi, H.; and Brown, M. 2019. Measuring the Effects of Non-Identical Data Distribution for Federated Visual Classification. arXiv:1909.06335.

Ilharco, G.; Ribeiro, M. T.; Wortsman, M.; Gururangan, S.; Schmidt, L.; Hajishirzi, H.; and Farhadi, A. 2023. Editing Models with Task Arithmetic. arXiv:2212.04089.

Izmailov, P.; Podoprikhin, D.; Garipov, T.; Vetrov, D. P.; and Wilson, A. G. 2018. Averaging Weights Leads to Wider Optima and Better Generalization. In *Proceedings of the 34th Conference on Uncertainty in Artificial Intelligence*, 876–885.

Jang, D.-H.; Yun, S.; and Han, D. 2025. Model Stock: All we need is just a few fine-tuned models. arXiv:2403.19522.

Jin, X.; Ren, X.; Preotiuc-Pietro, D.; and Cheng, P. 2025. Dataless Knowledge Fusion by Merging Weights of Language Models. arXiv:2212.09849.

Kairouz, P.; McMahan, H. B.; Avent, B.; et al. 2021. Advances and Open Problems in Federated Learning. *Foundations and Trends in Machine Learning*, 14(1–2): 1–210.

Kaissis, G. A.; Makowski, M. R.; Rückert, D.; and Braren, R. F. 2020a. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence*, 2(6): 305–311.

Kaissis, G. A.; Makowski, M. R.; Rückert, D.; and Braren, R. F. 2020b. Secure, privacy-preserving and federated machine learning in medical imaging. *Nature Machine Intelligence*, 2(6): 305–311.

Karimireddy, S. P.; Kale, S.; Mohri, M.; Reddi, S.; Stich, S.; and Suresh, A. T. 2020. SCAFFOLD: Stochastic Controlled Averaging for Federated Learning. In *Proceedings of the 37th International Conference on Machine Learning*, 5132–5143.

Li, T.; Sahu, A. K.; Talwalkar, A.; and Smith, V. 2020. Federated Optimization in Heterogeneous Networks. In *Proceedings of Machine Learning and Systems*, volume 2, 429–450.

Li, X.; Jiang, M.; Zhang, X.; Kamp, M.; and Dou, Q. 2021. Federated Learning on Non-IID Features via Local Batch Normalization. In *International Conference on Learning Representations*.

Liu, Z.; Mao, H.; Wu, C.-Y.; Feichtenhofer, C.; Darrell, T.; and Xie, S. 2022. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 11976–11986.

Liu, Z.; et al. 2021. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 10012–10022.

Marczak, D.; Magistri, S.; Cygert, S.; Twardowski, B.; Bagdanov, A. D.; and van de Weijer, J. 2025. No Task Left Behind: Isotropic Model Merging with Common and Task-Specific Subspaces. arXiv:2502.04959.

Matena, M.; and Raffel, C. 2022. Merging Models with Fisher-Weighted Averaging. arXiv:2111.09832.

McMahan, B.; Moore, E.; Ramage, D.; Hampson, S.; and Arcas, B. A. y. 2017. Communication-Efficient Learning of Deep Networks from Decentralized Data. In Singh, A.; and Zhu, J., eds., *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, 1273–1282. PMLR.

Rieke, N.; Hancox, J.; Li, W.; Milletarì, F.; Roth, H. R.; Albarqouni, S.; Bakas, S.; Galtier, M. N.; Landman, B. A.; Maier-Hein, K.; Ourselin, S.; Sheller, M.; Summers, R. M.; Trask, A.; Xu, D.; Baust, M.; and Cardoso, M. J. 2020. The future of digital health with federated learning. *npj Digital Medicine*, 3(1): 171.

Sheller, M. J.; Edwards, B.; Reina, G. A.; Martin, J.; Pati, S.; Kotrotsou, A.; Milchenko, M.; Xu, W.; Marcus, D.; Colen, R. R.; and Bakas, S. 2020. Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data. *Scientific Reports*, 10(1): 10933.

Shukla, S. N.; and Marlin, B. M. 2021. Multi-Time Attention Networks for Irregularly Sampled Time Series. arXiv:2101.10318.

Stoica, G.; Bolya, D.; Bjorner, J.; Ramesh, P.; Hearn, T.; and Hoffman, J. 2024. ZipIt! Merging Models from Different Tasks without Training. In *International Conference on Learning Representations*.

van der Maaten, L.; and Hinton, G. 2008. Visualizing Data Using t-SNE. *Journal of Machine Learning Research*, 9(86): 2579–2605.

Warnat-Herresthal, S.; Schultze, H.; Shastry; and so on. 2021. Swarm Learning for decentralized and confidential clinical machine learning. *Nature*, 594(7862): 265–270.

Wei, Y.; Zhu, Q.; Xia, Z.; Wang, L.; Zou, Y.; Li, Y.; Luo, Y.; and Du, B. 2026. MedMerge: A Training-Free Model Merging Framework for Medical Knowledge Transfer into Vision–Language Models. *IEEE Transactions on Radiation and Plasma Medical Sciences*, 1–1.

Wortsman, M.; Ilharco, G.; Gadre, S. Y.; Roelofs, R.; Gontijo-Lopes, R.; Morcos, A. S.; Namkoong, H.; Farhadi, A.; Carmon, Y.; Kornblith, S.; and Schmidt, L. 2022. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In Chaudhuri, K.; Jegelka, S.; Song, L.; Szepesvari, C.; Niu, G.; and Sabato, S., eds., *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, 23965–23998. PMLR.

Yadav, P.; Tam, D.; Choshen, L.; Raffel, C.; and Bansal, M. 2023. TIES-Merging: Resolving Interference When Merging Models. arXiv:2306.01708.

Yang, J.; Shi, R.; and Ni, B. 2023. MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. *Scientific Data*, 10(1): 41.

Yu, L.; Yu, B.; Yu, H.; Huang, F.; and Li, Y. 2024. Language Models are Super Mario: Absorbing Abilities from Homologous Models as a Free Lunch. arXiv:2311.03099.

Zech, J. R.; Badgeley, M. A.; Liu, M.; Costa, A. B.; Titano, J. J.; and Oermann, E. K. 2018. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, 15(11): 1–17.

Zeng, F.; Guo, H.; Zhu, F.; Shen, L.; and Tang, H. 2025. RobustMerge: Parameter-Efficient Model Merging for MLLMs with Direction Robustness. arXiv:2502.17159.

Zhang, X.; Zeman, M.; Tsiligkaridis, T.; and Zitnik, M. 2022. Graph-Guided Network for Irregularly Sampled Multivariate Time Series. arXiv:2110.05357.

Zheng, S.; and Wang, H. 2025. FREE-Merging: Fourier Transform for Efficient Model Merging. arXiv:2411.16815.

Supplementary Results and Analysis

Evaluation Datasets

The five MedMNIST v2 benchmarks(Yang, Shi, and Ni 2023) cover blood-cell microscopy, dermoscopy, abdominal CT organ slices, and ultrasound images. They involve cell morphology recognition, skin-lesion diagnosis, organ-structure recognition, and ultrasound-structure discrimination:

- **Blood**: an eight-class blood-cell microscopy dataset for hematological morphology classification, focusing on fine-grained cellular appearance and inter-class morphological similarity.
- **Derma**: a seven-class dermoscopic skin-lesion dataset, characterized by strong long-tailed prevalence and visually subtle lesion categories.
- **Organ-C**: an 11-class abdominal CT dataset in the coronal view, evaluating organ-category recognition under grayscale cross-sectional imaging.
- **Organ-S**: an 11-class abdominal CT dataset in the sagittal view, complementing Organ-C with a different anatomical imaging plane.
- **Ultrasound**: an eight-class ultrasound image dataset, covering structure recognition under modality-specific speckle noise and low-contrast boundaries.

Together, these benchmarks provide a systematic testbed for evaluating cross-modality robustness, class-discriminative capability, and task generalization under different medical image types and class-distribution conditions.

Detailed Ablation Configurations

All diagnostic-prototype substitutions retain long-tail prevalence calibration and vary only prototype semantics or support-count weighting:

- **LAMP-Merge** uses $p_c = \sum_i \alpha_{i,c}\mu_{i,c}$, where $e_{i,c} = (n_{i,c}+1)^\gamma \mathbf{1}[n_{i,c}>0]$ determines the class-support reliability.
- **Classifier-head aggregation** replaces the reference-space class prototype by the local classifier-head direction, i.e., $\mu_{i,c} \leftarrow h_{i,c}$.
- **Global-feature mean** removes class conditioning by setting $\mu_{i,c} \leftarrow g$ for all c, where g is the class-agnostic global reference-feature mean.
- **Support-only synthetic head** removes image evidence by replacing each prototype with a fixed random unit direction u_c, i.e., $p_c \leftarrow u_c$, $\|u_c\|_2 = 1$.
- **Shuffled-label prototype** preserves prototype magnitudes but permutes diagnostic semantics, i.e., $p_c \leftarrow p_{\sigma(c)}$, where σ is a random permutation over diagnosis labels.
- **Uniform client weight** replaces support-weighted aggregation with equal weights among clients containing class c, i.e., $\alpha_{i,c} \leftarrow \mathbf{1}[n_{i,c}>0]/\sum_j \mathbf{1}[n_{j,c}>0]$.
- **Binary support only** replaces fine-grained support and prevalence counts with class-presence indicators, i.e., $n_{i,c}, m_{i,c} \leftarrow \mathbf{1}[n_{i,c}>0]$.

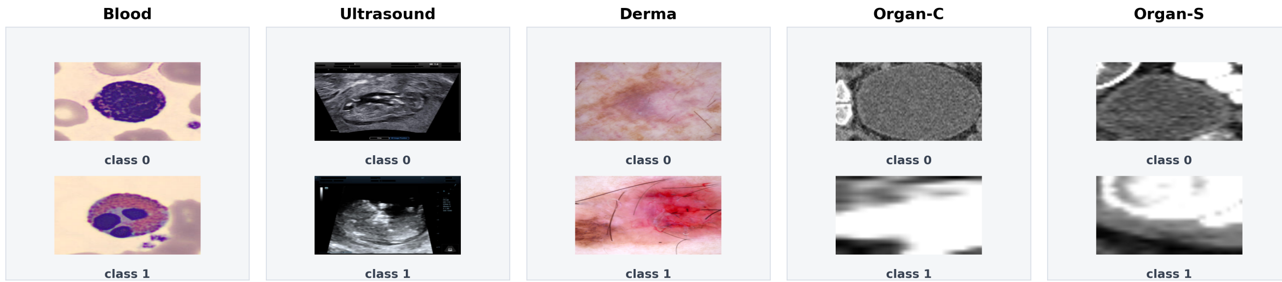

Figure 6: Representative training images from the five medical image datasets used in the experiments. The class IDs are shown only to illustrate category diversity within each dataset.

- **Global client-size weight** uses the total client support $N_i = \sum_k n_{i,k}$ rather than class-specific support, i.e., $e_{i,c} \leftarrow (N_i + 1)^\gamma \mathbf{1}[n_{i,c} > 0]$.

The long-tail prevalence substitutions vary only the class prior used by the calibration bias:

- **LAMP-Merge** estimates the empirical class prior as $\pi_c = \sum_i m_{i,c} / \sum_k \sum_i m_{i,k}$.
- **No prevalence calibration** removes the calibration bias by setting $b_c \leftarrow 0$.
- **Uniform prevalence prior** removes long-tail prevalence by setting $\pi_c \leftarrow 1/C$, where C is the number of diagnosis classes.
- **Smoothed prevalence prior** uses $\pi_c \leftarrow \sum_i (m_{i,c} + 1) / \sum_k \sum_i (m_{i,k} + 1)$, preserving the same prior structure with additive smoothing.

2) Additional Theoretical Analysis Let $\mu_c^\star$ denote the true class mean in the shared reference space. If client class features have bounded second moments, the prototype estimation error satisfies

$$\mathrm{E}\|p_c - \mu_c^\star\|_2^2 \leq \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \mathrm{Bias}_c^2. \tag{10}$$

This upper bound shows that support counts directly control the variance of class-prototype estimation. A client that does not observe class c should not contribute evidence for that class, and a client with more samples should contribute more, but only sublinearly.

Let the true margin between classes c and d for sample x be

$$\Delta_{c,d}(x) = (\mu_c^\star)^\top z - (\mu_d^\star)^\top z. \tag{11}$$

When the prototype error satisfies

$$\|p_c - \mu_c^\star\|_2 + \|p_d - \mu_d^\star\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \tag{12}$$

the estimated prototypes preserve the class order between c and d.

The role of long-tail prevalence calibration is to introduce long-tail priors in a bounded manner. For prevalence counts $m_{i,c}$, the class prior is

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \tag{13}$$

and the centered log-prior bias is

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \tag{14}$$

As long as λ is bounded, long-tail prevalence calibration does not replace the discriminative directions reconstructed by diagnostic prototype reconstruction; it only calibrates the output toward the real class prevalence. This explains why using long-tail prevalence calibration alone is insufficient, while full LAMP-Merge remains both non-collapsed and prevalence-aware.

Metric Definitions and Geometry Details

Let y_n and $\hat{y}_n$ denote the true and predicted labels of the n-th test sample. Let $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$ be the predicted-class distribution, $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$ the true class distribution, and $\mathcal{C}_{\text{test}} = \{c \mid \sum_n \mathbf{1}[y_n = c] > 0\}$ the set of classes observed in the test set. We use:

- $\mathrm{BA} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \mathrm{Recall}_c$, balanced accuracy or macro recall, which measures whether all observed diagnosis classes are recalled.
- $\mathrm{MacroF1} = |\mathcal{C}_{\text{test}}|^{-1} \sum_{c \in \mathcal{C}_{\text{test}}} \mathrm{F1}_c$, class-balanced precision-recall quality that penalizes majority-only predictions.
- $\rho = \max_c q_c$, the prediction-collapse ratio, which measures concentration on a single diagnosis class.
- $C_{\text{eff}} = \exp\left(-\sum_c q_c \log q_c\right)$, the effective number of predicted classes.
- $\mathrm{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|$, the total variation distance between predicted and true class distributions.

Let $d(a, b) = 1 - \cos(a, b)$ denote cosine distance and let $\mathcal{I}_c = \{i \mid n_{i,c} > 0\}$ denote the clients that contain diagnosis class c. The geometric quantities are:

- $D_{\text{pair}} = \frac{2}{C(C-1)} \sum_{c<d} d(p_c, p_d)$, the mean distance between global prototypes of different diagnosis classes.
- $D_{\text{nn}} = \frac{1}{C} \sum_c \min_{d \neq c} d(p_c, p_d)$, the mean nearest-class distance and most vulnerable decision margin.
- $A_{\text{client}} = \frac{1}{C} \sum_c \mathrm{Avg}_{i<j,\, i,j \in \mathcal{I}_c} \cos(\mu_{i,c}, \mu_{j,c})$, the within-class consistency among client prototypes; $A_{\text{proto}} = \frac{1}{C} \sum_c \sum_{i \in \mathcal{I}_c} \alpha_{i,c} \cos(p_c, \mu_{i,c})$, the alignment between the global prototype and contributing client prototypes.